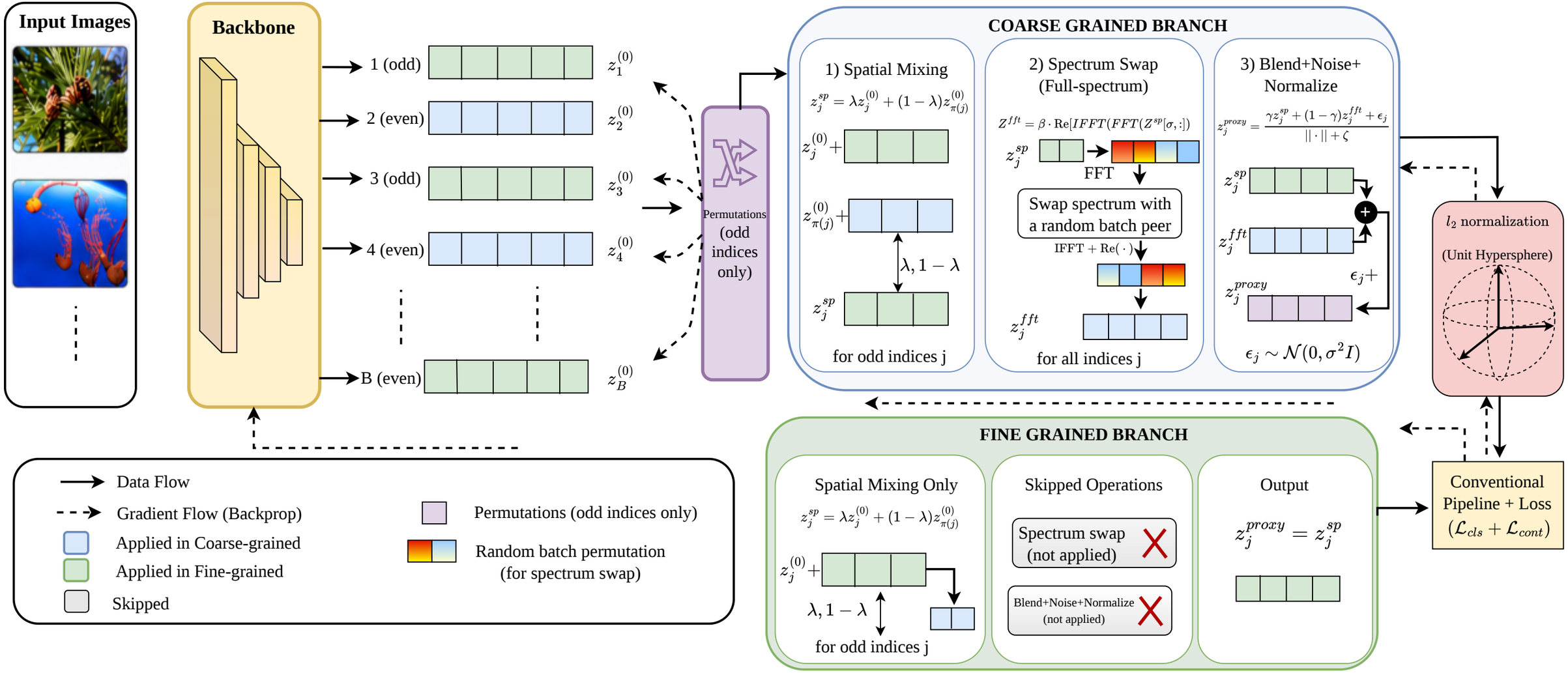
Input Images
Backbone
1 (odd)
2 (even)
3 (odd)
4 (even)
B (even)
$z_1^{(0)}$
$z_2^{(0)}$
$z_3^{(0)}$
$z_4^{(0)}$
$z_B^{(0)}$
Permutations (odd indices only)
COARSE GRAINED BRANCH
1) Spatial Mixing
$z_j^{sp} = \lambda z_j^{(0)} + (1-\lambda) z_{\pi(j)}^{(0)}$
$z_j^{(0)} +$
$z_{\pi(j)}^{(0)} +$
$\lambda, 1-\lambda$
z_j^{sp}
for odd indices j
2) Spectrum Swap (Full-spectrum)
$Z^{fft} = \beta \cdot \mathrm{Re}[IFFT(FFT(Z^{sp}[\sigma,:])$
z_j^{sp}
FFT
Swap spectrum with a random batch peer
IFFT + Re(·)
z_j^{fft}
for all indices j
3) Blend+Noise+ Normalize
$z_j^{proxy} = \frac{\gamma z_j^{sp} + (1-\gamma) z_j^{fft} + \epsilon_j}{||\cdot|| + \zeta}$
z_j^{sp}
z_j^{fft}
$\epsilon_j +$
z_j^{proxy}
$\epsilon_j \sim \mathcal{N}(0, \sigma^2 I)$
l_2 normalization
(Unit Hypersphere)
FINE GRAINED BRANCH
Spatial Mixing Only
$z_j^{sp} = \lambda z_j^{(0)} + (1-\lambda) z_{\pi(j)}^{(0)}$
$z_j^{(0)} +$
$\lambda, 1-\lambda$
for odd indices j
Skipped Operations
Spectrum swap (not applied)
Blend+Noise+Normalize (not applied)
Output
$z_j^{proxy} = z_j^{sp}$
Conventional Pipeline + Loss
$(\mathcal{L}_{cls} + \mathcal{L}_{cont})$
Data Flow
Gradient Flow (Backprop)
Applied in Coarse-grained
Applied in Fine-grained
Skipped
Permutations (odd indices only)
Random batch permutation (for spectrum swap)